Ziphozinhle Ngqopi

57/60
95%

ELECTRICITY AND ELECTRONICS

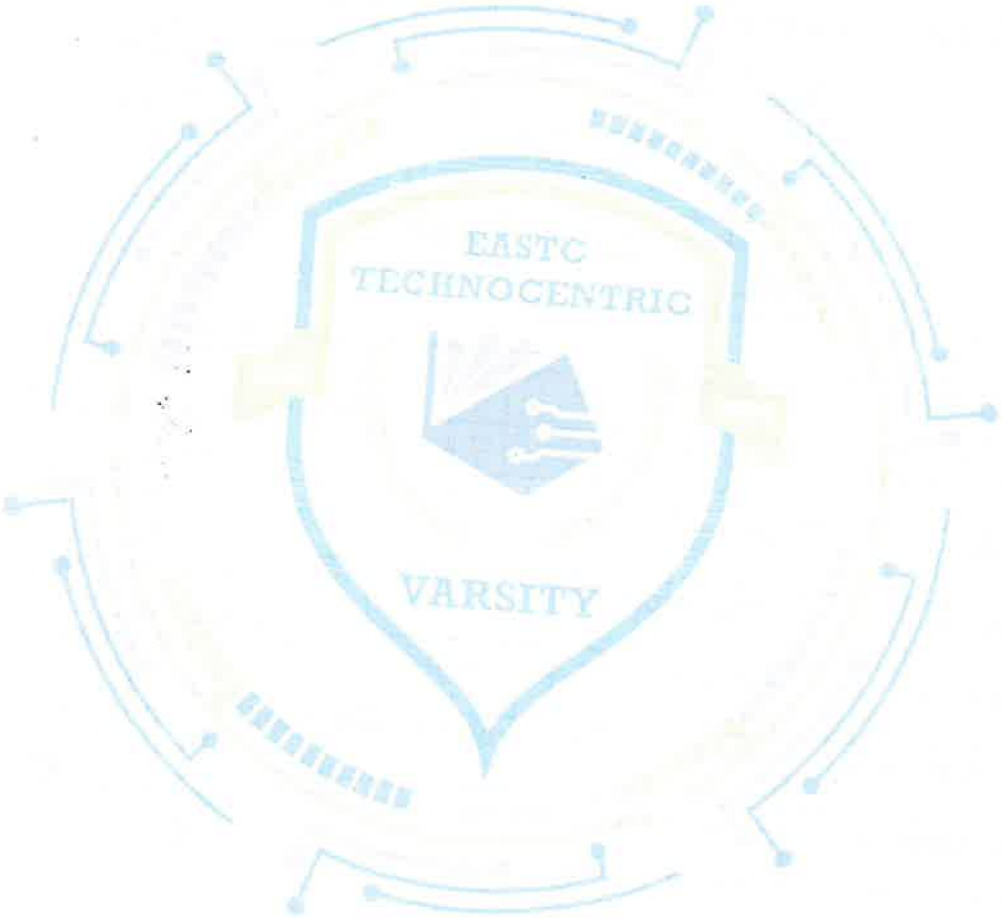
SUMMATIVE ASSESMENTS

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).



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Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Ziphozinle Maqopi	Manager:
Trainer:	Mentor / Coach:
Assessor: Baleenah Jabadi	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozinle Maqopi

Learner's signature: _____ Date: _____

Assessors Name: Baleenah Jabadi

Assessor's signature: _____ Date: _____

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

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Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

EASTC TECHNOCENTRIC VARSITY

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Zipnozihle Maqopi	Signature
Assessors Name:	Signature:

EASTC TECHNOCENTRIC VARSITY

LEARNING UNIT 1:
KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

Task	Questions Description	Marks
1	List types of materials used for conductors, insulators and semi-conductors and describe their mechanical and electrical properties and applications.	10

Conductors

- Metals: Copper, Silver, Aluminium, Gold
- Non-metals: Graphite
- Liquids: Salt water, acidic solutions

Electrical properties

- High electrical conductivity due to free electrons
- Low resistivity
- Linear voltage-current relationship
- Conductivity slightly affected by temperature

Mechanical properties

- Metals are ductile and malleable allowing wires to be drawn or bent
- Corrosion resistance varies (Gold and Aluminium resist corrosion; copper may oxidize)

Application

- Electrical wiring - Copper Aluminium

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Task	Questions Description	Marks
2	a) Name, describe and explain the different types of magnets and their properties b) Explain fundamental magnetic concepts by naming the five characteristics of magnetic lines of force and explaining the relationship between flux and flux density	10

2a)

1) Neodymium Magnets

strongest type of peimant magnet, It is a high energy product. Brittle, prone to corrosion and ~~sense~~ sensitive to heat. Used in motors, generators, speakers and magnetic levitation system

2.) Ferrite Magnets

Most common type of permanent magnet. Inexpensive, resistant to corrosion, and stable. Lower energy produca compared to Neodymium magnets. Used in speakers, motors and magnetic separators

3.) Alnico Magnets

Made from aluminium, nickel and cobalt. Good temperature stability and corrosion resistance. Medium strength can be brittle. Used in electric motors, sensors

4.) Samarium - Cobalt Magnets

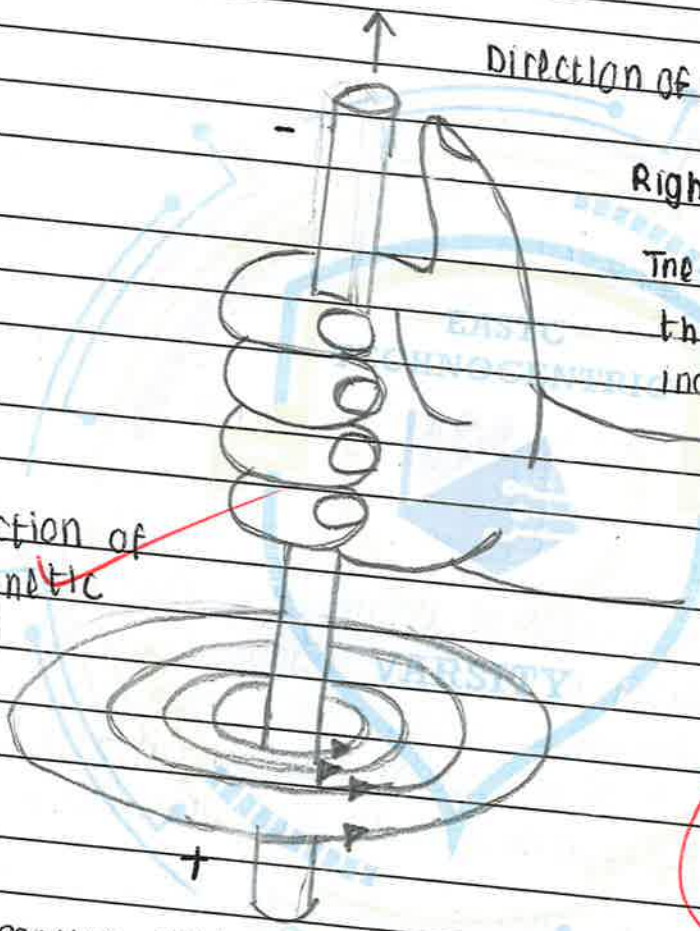
High energy product, similar to ~~N~~ Neodymium magnets. It has excellent temperature stability and corrosion resistance. It is used in high-temperature applications, aerospace, and precision motors

5) Electromagnets

Temporary magnets created by electric current. It can be turned on/off and adjusted. Often used with iron cores to enhance field strength. Used in electric motors, relays, magnetic levitation

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Task	Questions Description	Marks
3	Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor.	10



Right-Hand Rule

The arrows around the conductor indicate the circular magnetic field, following the curl of the fingers

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Mag Magnetic field Establishment

When an electric current flows through a conductor, it generates a magnetic field around it. The strength of the magnetic field (B) at a distance is given by:

$$B = \frac{(\mu_0 I)}{2\pi r}$$

μ_0 = permeability of free space ($4\pi \times 10^{-7}$ T·m/A)

I = current in amperes

r = distance from the conductor in meters

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standards.

Earthing, also known as grounding, is the process of connecting the electrical system to the ground to ensure safety. This provides a low-impendance path for fault currents, allowing protective devices to operate quickly and prevent electrical shock. The earthing system typically consists of an earth electrode such as a rod or plate, connected to the electrical system's neutral point.

Bonding: Bonding is the practice of connecting various conductive parts to ensure they have the same electrical pontential energy reducing the risk of electrical shock

Regulatory Requirements

~~• National Electrical Code (NEC)~~

• ~~Nation~~ • International Electiotechinical Commision: IEC Standards such as IEC 60364, provide guidelines for earthing and bonding in electrical systems globally

• Local Regulations: Always check local codes and standards, as they may have specific requiremems that differ from national or intemational standards.

Compliance with earthing and bonding regulations is crucial to ensure safety and prevent electrical shock hazards and equipment damage.

LEARNING UNIT 3:
KM-03-KT02: Electronics (30%)

Learning Outcome 1:	International Electro technical Commission (IEC) symbols for electronic components
Learning Outcome 2:	Principles, safety precautions, identification and basic function of electronic components
Learning Outcome 3:	Basic electronic circuits
Learning Outcome 4:	Principles of rectification, single phase and 3-phase

Task	Questions Description	Marks
6	a) Describe the various inductors used and typical applications b) Describe the construction and uses of various semiconductor devices c) Draw basic electronic circuits using International Electrotechnical Commission (IEC) symbols	10

ba). Air core Inductors: These inductors do not have a magnetic core, instead, they use air as a medium. They are used in RF amplifiers, antennas, and oscillators

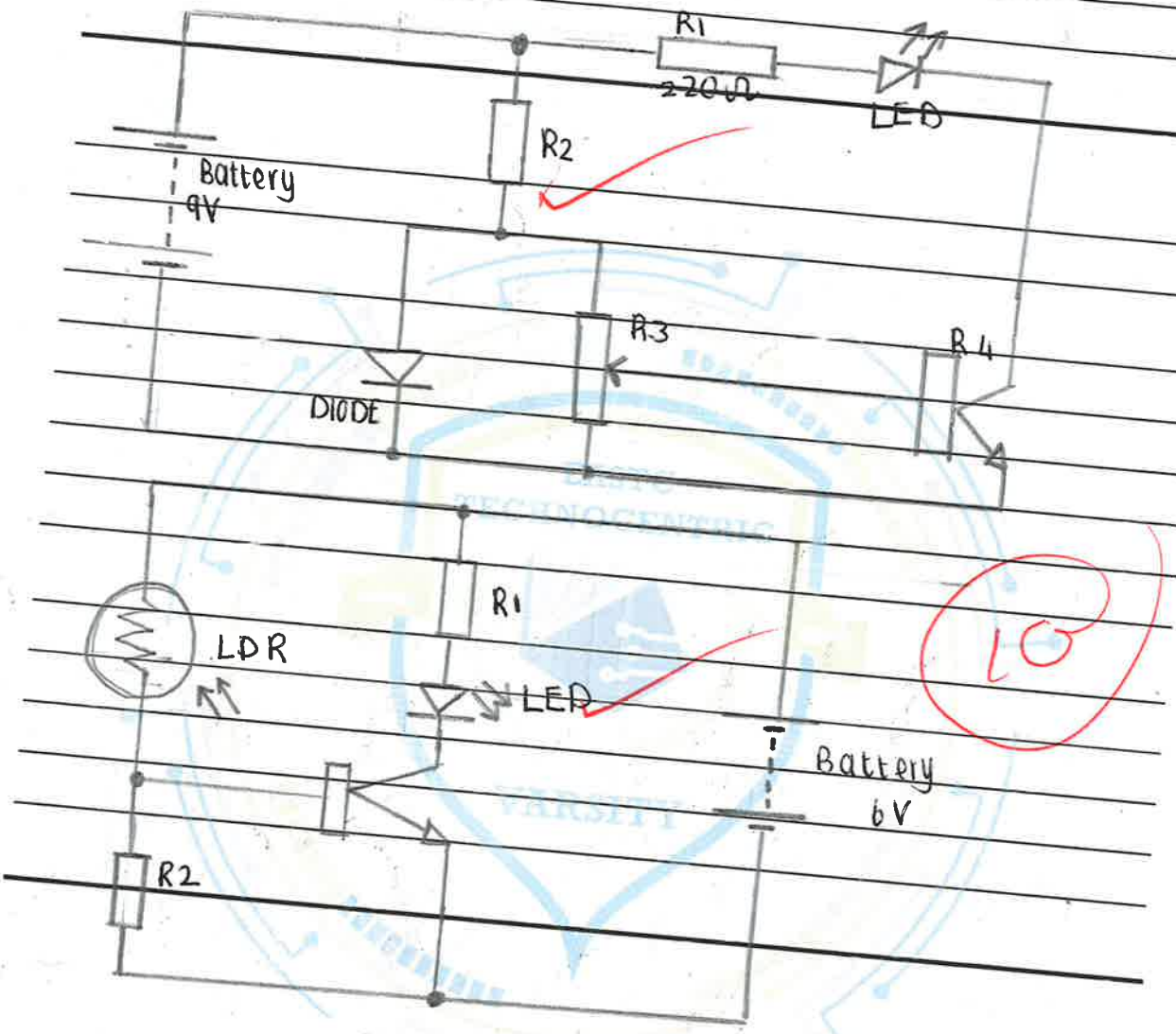
Iron Core Inductors: These inductors use and iron to increase inductance and improve efficiency. They are commonly used in power transformers and inductors for low-frequency applications

Ferrite Core Inductors: These inductors are made from a ceramic material that is magnetically conductive. They

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recombine they release photons. LEDS are used in displays, lighting and as indicators

c)



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Assessor's Report

Candidate's Name	Ziphozinle Mqopi	ID No.	0802030118086
Assessor's Name	~~EB~~ Balcraigh T	Reg. No.	NAMB - A1 2025/5307
Title	Electricity and Electronics		

ASSESSMENT DECISION

Outcome	C	NYC	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments

~~Good work~~

Date	
~~EB~~	
Signature of Assessor	Signature of Candidate

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Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

EASTC TECHNOCENTRIC VARSITY

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ASSESSMENT REVIEW

NAME of LEARNER Zipnozinle Maqoqi	NAME of ASSESSOR Balenah Yabadi
VENUE EASTC TECNOCENTRIC VARSITY	DATE of REVIEW

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment was practical?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The venue and equipment was functional?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
An opportunity to appeal was given	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The evidence was recorded	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING

I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid

Learner	Date	Assessor	Date	Moderator	Date